

Project 3: Skin Lesion Classification for Melanoma Detection

Develop a deep learning model to classify skin lesion images as either **benign** (non-cancerous) or **malignant** (cancerous). This is a critical task in computer-assisted dermatology.

- **Core Concepts:** Image Classification, Convolutional Neural Networks (CNNs), Transfer Learning, TensorFlow/PyTorch.

Dataset

You'll use the **HAM10000 ("Human Against Machine with 10000 training images")** dataset. It's a large collection of multi-source dermatoscopic images of common pigmented skin lesions.

- **Instructions:**
 1. Download the dataset from the [Harvard Dataverse](#).
 2. You will need the image files and the HAM10000_metadata.csv file.
 3. The 'dx' column in the metadata file contains the diagnosis. You will map these diagnoses into two classes: 'benign' and 'malignant'. (e.g., 'mel' and 'bcc' are malignant; others like 'nv' and 'bkl' are benign).

Starter Code

This snippet (provided separately) demonstrates how to set up a basic image classification model using **TensorFlow/Keras** with transfer learning from MobileNetV2.

Overall Guidance

1. **Data Preparation:** Load the metadata and create a balanced dataset. The original dataset is highly imbalanced. Use techniques like over-sampling the minority class (malignant) or under-sampling the majority class (benign).
2. **Image Augmentation:** Use ImageDataGenerator (Keras) or Transforms (PyTorch) to apply random rotations, flips, and zooms to the training images. This helps the model generalize better and prevents overfitting.
3. **Transfer Learning:** Use a pre-trained CNN like **MobileNetV2**, **ResNet50**, or **EfficientNet**. Freeze the initial layers and add your own classification head (a few Dense layers).
4. **Training & Evaluation:** Train the model and monitor its performance on a validation set. Evaluate the final model on a held-out test set using metrics like **Accuracy**, **Precision**, **Recall**, and the **AUC-ROC curve**. The ROC curve is particularly important for medical diagnostic tests.

Evaluation Rubric (100 points)

Criteria (Total 100)	Poor (0-9)	Fair (10-14)	Good (15-20)
Translate business problem into technical problem (10)			
Data Handling (10)	Fails to address class imbalance.	Attempts to balance data but method is flawed.	Effectively handles class imbalance; uses data augmentation properly.
Model Architecture (20)	Incorrect model implementation.	A basic CNN is used without transfer learning.	Correctly implements a transfer learning model with a proper architecture.
Performance & Evaluation (15)	Low accuracy and poor AUC.	Moderate performance; evaluation is limited to accuracy.	High accuracy and AUC; comprehensive evaluation with ROC curve analysis.
Code & Report (15)	Disorganized code, no report.	Functional code, minimal documentation.	Clean, well-documented code with a clear report explaining the methodology.
Holistic approach (10)			
Team Work (10)			
Presentation (10)			